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Roll No

EE-701 (GS)**B.E. VII Semester** Examination, June 2020**Grading System (GS)****Power System Analysis and Control****Time : Three Hours****Maximum Marks : 70**

- Note:** i) Attempt any five questions.
ii) All questions carry equal marks.

1. A 100 MVA synchronous generator operates on full load at a frequency of 50Hz. The load is suddenly reduced to 50 MW. Due to time lag in governor system, the valve begins to close after 0.45. Determine the change in frequency that occurs in this time. Given $H = 5 \text{ kWS/kVA}$.
2. What do you understand by restructuring process? Describe various entities involved in restructuring.
3. Discuss the term 'dynamic incremental state variables and coherency'. Briefly outline the power control mechanism of an individual generator and develop the mathematical model of a speed governing system. Derive transfer functions.
4. Explain how Gauss-Seidel iterative technique is applied to
 - a) Load buses
 - b) Voltage controlled buses.
5. What is equal area criterion. Discuss its application and limitation in the study of power system stability.
6. The following is the system data for a load flow solution. The line admittances.

Bus code	Admittance
1-2	-j 5.0
1-3	-j 5.0
1-4	-j 5.0
2-3	-j 10.0
3-4	-j 10.0

The schedule of Active and Reactive Powers

Bus code	P	Q	V	Remarks
1	-	-	1+j0.0	Slack bus
2	1	0.1	-	PQ Bus
3	3.5	0.3	-	PQ Bus
4	1	0.2	-	PQ Bus

Find the voltage at the end of the first iteration using Gauss- Seidel method.

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7. Describe the methods for solving economic dispatch using Lagrangian multiplier method.

OR

Discuss the challenges and problems faced by Indian Power Industry.

8. Write short notes on (Any two):

- a) Economic dispatch
- b) Deregulation
- c) Voltage control in power system
- d) Excitation system
